Md. Jubair Uddin Zidan

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Academic Credentials

B.Sc in Mechanical Engineering, **2019 Feb - 2024 November**
Chittagong University of Engineering & Technology (CUET) Chittagong, Bangladesh
CGPA : 3.29/4.00

M.Sc in Mechanical Engineering, **2024 January - Early 2027** (Expected)
Chittagong University of Engineering & Technology (CUET) Chittagong, Bangladesh
CGPA : 3.90.../4.00

Professional Experience

Internship/Industrial Trainee, *RPCL Raozan 25 MW Dual Fuel Power Plant* **2023 Sept**
Completed a 3-week internship to acquire knowledge in power generation process Chittagong

Standardized Test Scores

IELTS : 7.0
Listening: 8.5 Reading: 7 Listening: 6.5 Listening: 6.5

GRE : To be taken soon

Research Experience

Computational Aerodynamics & Ground-Effect Modeling
Analyzed airflow behavior on pre-2022 vs. 2022 F1 chassis using ANSYS Fluent with k-ω SST modeling. Built CAD geometry, generated high-resolution meshes, and quantified changes in drag, downforce, velocity fields, and wake structure introduced by Venturi-based ground-effect tunnels.

Jet Noise Reduction & Chevron Nozzle Acoustics
Designed serrated nozzles in SolidWorks and performed acoustic simulations in ANSYS Fluent. Compared N12 vs N16 chevron configurations using contours of acoustic power, static pressure, and jet-mixing behavior to determine optimal noise-suppressing geometry.

Hybrid Vertical Axis Wind Turbine (VAWT) Performance
Modeled a Darrieus–Savonius hybrid VAWT in 3D and carried out transient CFD in ANSYS CFX. Fabricated prototype, executed wind-tunnel testing, and validated CFD predictions by correlating voltage, current, and power output.

Airfoil Optimization for VAWT Blades
Evaluated multiple symmetric and cambered airfoils for VAWT blades using CFD simulations across varying tip-speed ratios. Analyzed pressure fields, torque behavior, and power coefficients to assess airfoil influence on turbine efficiency.

Acoustic Attenuation in Gas-Turbine Combustor Liners
Developed double cylindrical perforated-liner geometry in SolidWorks and simulated acoustic damping using ANSYS Fluent PVPM (porous-media). Assessed acoustic power reduction, noise distribution, and liner effectiveness compared to conventional designs.

Publications

Conference Proceedings

[1] S. I. Uddin, G. M. S. Rayhan, T. Abdullah, **M. J. U. Zidan**, and B. Roy, "Numerical and experimental performance analysis of a Darrieus–Savonius hybrid vertical axis wind turbine for low wind speed conditions," in Proc. 8th Int. Conf. Mechanical, Industrial and Energy Engineering (ICMIEE), 2024.

[2] M. Hassan, **M. J. U. Zidan**, and M. R. Zaman, "Modelling and CFD acoustic analysis of N12 and N16 chevron nozzles for jet noise reduction," in Proc. Int. Conf. Mechanical, Industrial and Materials Engineering (ICMIME), 2024.

[3] M. Hassan and **M. J. U. Zidan**, "Noise emission reduction of a jet engine by double cylindrical perforated liners in a combustion chamber," in Proc. Int. Conf. Mechanical, Industrial and Materials Engineering (ICMIME), 2024.

[4] M. K. I. Monna, T. Nath, and **M. J. U. Zidan**, "A numerical approach for evaluating the aerodynamic effectiveness of vertical axis wind turbine by different types of airfoils," in Proc. 6th Int. Conf. Mechanical, Industrial and Materials Engineering (ICMIME), Rajshahi, Bangladesh, 2024.

Papers in progress

1. **M. J. U. Zidan, M Wakil,** Dibos Das, "Architected Auxetic Mechanical Metamaterials for Energy Absorption: Design, Fabrication, and Characterization"

Research Interest

Fluid Mechanics | Aerodynamics | Fluid Flow | Advanced Materials | Advanced Manufacturing

Undergraduate Thesis

A numerical approach for evaluating the aerodynamic effectiveness of venturi tunnel and ground effect on newer f1 cars

Supevisor: Prof. Dr. Sheikh Muhammad Humayun Kabir, Dept of Mechanical Engineering, CUET

Awards

1st Runner-Up, *Poster Presentation, Research Fair 2023*
Chittagong University of Engineering & Technology (CUET) 2023

Champion, *National Math Olympiad*
Barishal Region, Bangladesh 2011

2nd Runner-Up, *National Earth Olympiad*
Barishal Region, Bangladesh 2014

Champion, *Climate Justice Idea Competition, Season 2*
Powered by The Daily Star & ActionAid Bangladesh 2023

Certifications

Certified SolidWorks Associate (CSWA), *Dassault Systèmes* 2025

Certified SolidWorks Professional (CSWP), *Dassault Systèmes* 2026

Extracurricular Activities

Treasurer, *American Society of Mechanical Engineers (ASME), CUET Student Branch* 2023- 2024

President, *Supply Chain & Business Alliance CUET* 2023- 2024

Joint Publication Secretary, *CUET Automotive and Mobility Society (CAMS)* 2023- 2024

Technical Skills

CAD & Design: SolidWorks
Simulation : ANSYS Fluent, ANSYS LS-DYNA, LAMMPS, OpenFOAM (Foundational Knowledge), COMSOL Multiphysics (Foundational Knowledge)
Programming: MATLAB, Python, C (Coursework), Fortran (Coursework)
Scientific Computing & Data Analysis: NumPy, Pandas, Jupyter Notebook, OriginPro, MS Excel
Version Control: Git & GitHub
Scientific Documentation & Research: LaTeX, Mendeley, MS Word, MS PowerPoint, Google Suite
Graphic Design & Visualization: Adobe Illustrator, Adobe Photoshop, Canva, Figma, Adobe Premiere Pro
Operating Systems: macOS, Windows

References

1. **Prof. Dr. Sheikh Muhammad Humayun Kabir**, Professor, Dept of Mechanical Engineering, CUET
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2. **Prof. Dr. Md. Mamunur Roshid**, Professor, Dept of Mechanical Engineering, CUET
mamuncuet2003@cuet.ac.bd

3. **Prof. DR Jamal Uddin Ahamed**, Professor, Dept of Mechanical Engineering, CUET
jamal@cuet.ac.bd@cuet.ac.bd